

Circuits Description

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Summary

This document describes the classes of quantum circuits and respective data, available at <https://github.com/luisps/ExperimentalData.git>

Índice

Summary	1
Class SU - variational EfficientSU2	2
Circuit SU5: 5 qubits, 1 rep	2
Circuit SU15: 15 qubits, 2 rep.....	2
Circuit SU25: 25 qubits, 2 rep.....	3
Class HSC - Hidden Shift Circuits.....	3
Circuit 41256- Hidden shift: 256 qubits, hidden string = 1023.....	3
Circuit 411024- Hidden shift: 1024 qubits, hidden string = 1023	3
Circuit 414096- Hidden shift: 4096 qubits, hidden string = 1023	3
Class IQP Inversion test	3
Circuit IQP5: 5 qubits.....	3
Circuit IQP15: 15 qubits	4
Circuit IQP25: 25 qubits	4
Class RND - Random Circuits.....	4
Circuit RND8- Random: 8 qubits, nCycles = 6.....	4
Circuit RND10- Random: 10 qubits, nCycles = 7	5
Circuit RND512- Random: 12 qubits, nCycles = 8	5
Class RSQ - Square Random Circuits	5
Circuit RSQ4- Random square: 4 qubits, 4 layers	6
Circuit RSQ8- Random square: 8 qubits, 8 layers	6
Circuit RSQ12- Random square: 12 qubits, 12 layers	6
Circuit RSQ16- Random square: 16 qubits, 16 layers	6
Circuit RSQ18- Random square: 18 qubits, 18 layers	6

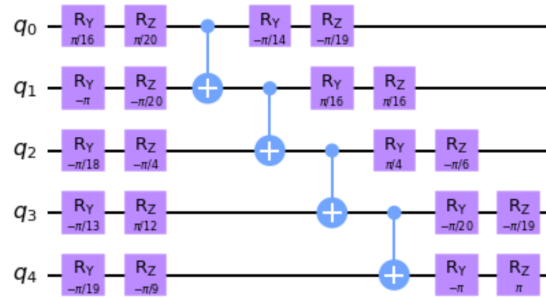
Class SU – variational EfficientSU2

This class of circuits corresponds to the so-called “hardware efficient” ansatz for variational circuits. It is obtained with the Qiskit method:

`EfficientSU2(num_qubits, entanglement='linear', reps=reps)`

The number of qubits and the number of layer repetitions is varied for the different circuit instances below.

Circuit SU5: 5 qubits, 1 rep

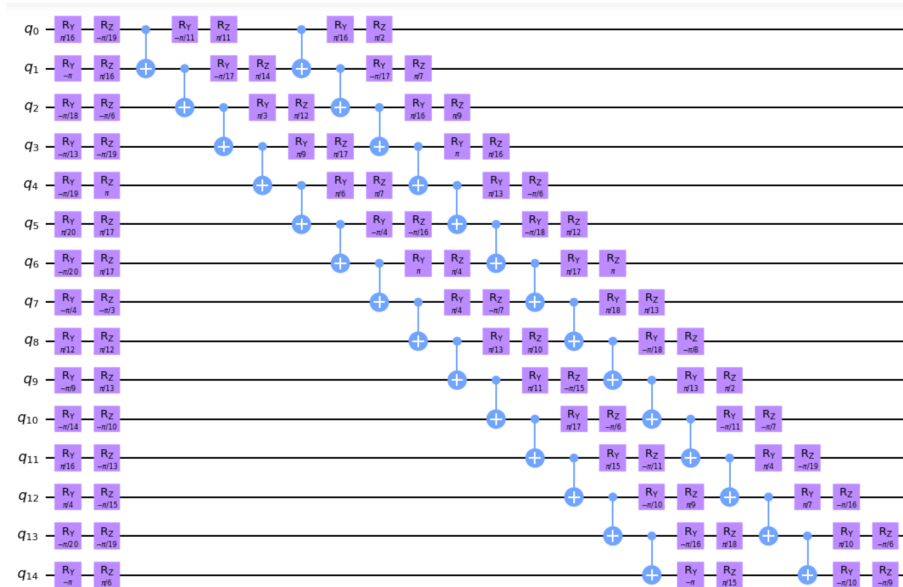


8 layers

$$3.436\text{e}+10 \text{ paths} = 2^{(5 \cdot (8-1))} = 2^{35}$$

$$\langle 0|U|0 \rangle = -0.00172150 + 0.00127863j$$

Circuit SU15: 15 qubits, 2 rep



22 layers

$$6.675\text{e}+94 \text{ paths} = 2^{(15 \cdot (22-1))} = 2^{315}$$

$$\langle 0|U|0 \rangle = 0.00364578 + 0.00060584j$$

Circuit SU25: 25 qubits, 2 rep

32 layers

$$2.e+233 \text{ paths} = 2^{(25*(32-1))}$$

$$\langle 0|U|0\rangle = -0.00012 + 0.00007j$$

Class HSC – Hidden Shift Circuits

Deterministic circuits, in the sense that only one basis state in the output has an amplitude different from 0 (= 1.0).

These are based on the algorithm (and code) presented in Peres, Filipa 2023 (sec 4.1)
<https://arxiv.org/pdf/2203.01789.pdf>

Circuit 41256- Hidden shift: 256 qubits, hidden string = 1023

43 layers

$$\text{paths} = 2^{(256*(43-1))}$$

$$\langle 1023|U|0\rangle = 1.0 + 0.0j$$

Circuit 411024- Hidden shift: 1024 qubits, hidden string = 1023

43 layers

$$\text{paths} = 2^{(1024*(43-1))}$$

$$\langle 1023|U|0\rangle = 1.0 + 0.0j$$

Circuit 414096- Hidden shift: 4096 qubits, hidden string = 1023

43 layers

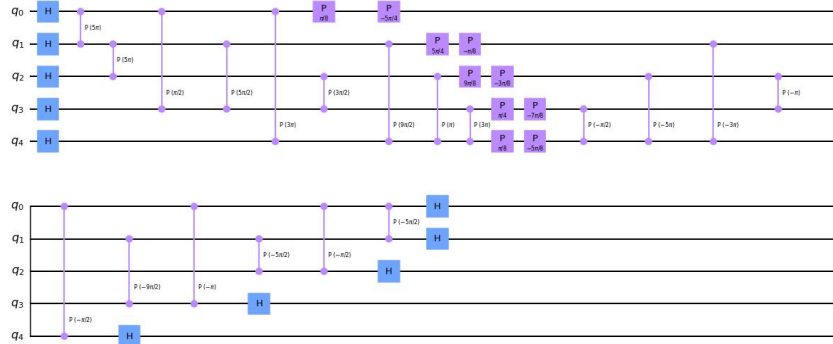
$$2^{(4096*(43-1))}$$

$$\langle 1023|U|0\rangle = 1.0 + 0.0j$$

Class IQP Inversion test

Inversion test of two random states encoded using IQP.
The probability of $P(|0\rangle^n)$ is the overlap between the states.

Circuit IQP5: 5 qubits



17 layers

$$1.209e+24 \text{ paths} = 2^{(5 \cdot (17-1))} = 2^{80}$$

$$\langle 0|U|0 \rangle = -0.104462 + 0.106694j$$

Circuit IQP15: 15 qubits

55 layers

$$6.828e+243 \text{ paths} = 2^{(15 \cdot (55-1))} = 2^{810}$$

$$\langle 0|U|0 \rangle = -0.00128171 - 0.00008649j$$

Circuit IQP25: 25 qubits

97 layers

$$\text{paths} = 2^{(25 \cdot (97-1))} = 2^{2400}$$

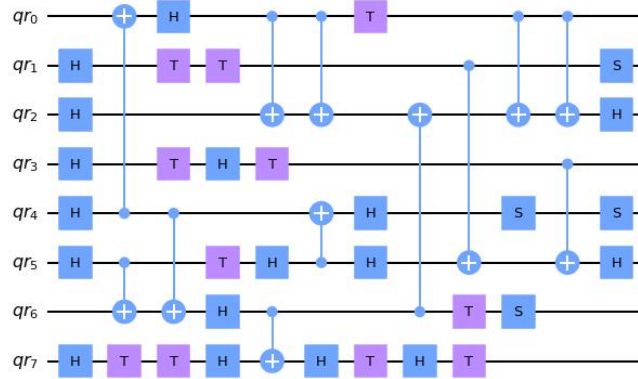
$$\langle 0|U|0 \rangle = -0.00013443 - 0.00022139j$$

Class RND – Random Circuits

Randomly generated circuits. These are based on the algorithm (and code) presented in Peres, Filipa 2023 (sec 4.2)

<https://arxiv.org/pdf/2203.01789.pdf>

Circuit RND8- Random: 8 qubits, nCycles = 6



9 layers

$$1.845e+19 \text{ paths} = 2^{(8 \times (9-1))} = 2^{64}$$

$$\langle 0|U|0 \rangle = -0.04419417 + 0.10669417j$$

Circuit RND10- Random: 10 qubits, nCycles = 7

13 layers

$$\text{paths} = 2^{(10 \times (13-1))} = 2^{120}$$

$$\langle 0|U|0 \rangle = 0.02209709 + 0.05334709j$$

Circuit RND512- Random: 12 qubits, nCycles = 8

15 layers

$$\text{paths} = 2^{(12 \times (15-1))} = 2^{168}$$

$$\langle 16|U|0 \rangle = 0.009153 - 0.022097j$$

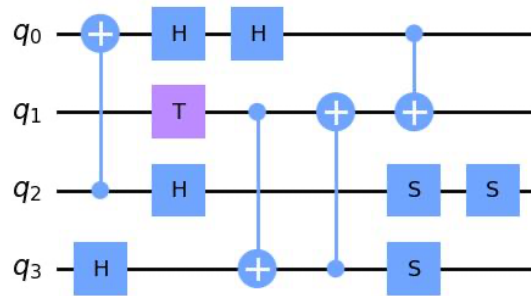
Class RSQ – Square Random Circuits

Randomly generated circuits with the same width (nbr qubits) and depth (nbr layers).

The circuits are generated by drawing gates from the Clifford+T gate set, with

Hadamard having a probability of 0.3.

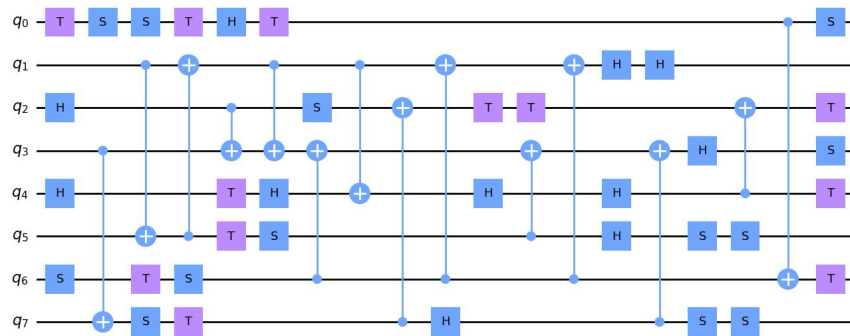
Circuit RSQ4- Random square: 4 qubits, 4 layers



$$\text{paths} = 2^{12}$$

$$\langle 0|U|0\rangle = 0.5+0.0j$$

Circuit RSQ8- Random square: 8 qubits, 8 layers



$$\text{paths} = 2^{56}$$

$$\langle 0|U|0\rangle = 0.15089+0.0625j$$

Circuit RSQ12- Random square: 12 qubits, 12 layers

$$\text{paths} = 2^{132}$$

$$\langle 0|U|0\rangle = -0.06440-0.02667j$$

Circuit RSQ16- Random square: 16 qubits, 16 layers

$$\text{paths} = 2^{240}$$

$$\langle 0|U|0\rangle = -0.00162-0.00391j$$

Circuit RSQ18- Random square: 18 qubits, 18 layers

$$\text{paths} = 2^{306}$$

$$\langle 0|U|0\rangle = 0.00138+0.00098j$$